Small Models, Certified Behavior

A runtime that lends domain knowledge to local LLMs on robots and appliances

The metis project*

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How to read the numbers. Every figure in this paper is taken from a committed document, result file or test of the `metis-agent` repository or its siblings, and is followed by a pointer [document § section; commit] to where it stands. Nothing was re-run for the paper. The experiments are the record of a design being fixed by measurement, so some of them report a configuration that was then rejected; the ruling that followed each one is stated with it. The two sibling papers are *Behavior Programs as Measures over Proofs* (the language, draft 0.5) and *Behavior Catalogs in Practice* (the catalog and the loop, in preparation).

Abstract

A small local language model — Gemma 4 E2B, 2.1 GB resident on a laptop GPU, or a CPU — cannot be trusted to hold a device's procedure, state or safety. We do not ask it to. `metis-agent` puts a *certified runtime* in charge: a behavior program from the catalog, in the language LLP, evaluated exactly in about ten milliseconds, owning state, procedure, probability and legality. The model gets narrow, checked jobs: map an utterance to one typed intent, narrate what happened, propose an explanation for a surprise, ask the person. Two agents share one kernel — a *domain agent* per loaded domain (a mapper, a replier, a narrator, the runtime, a host) and an *orchestrator* per device (the dialog, routing, the floor) — and the device gets smarter by catalog changes through a human-gated loop, never by retraining. We report the experiments that fixed the design: the model switch (E2B maps the 54-turn dog corpus 54/54 at 458 ms median, in 2136 MiB); intent as JSON text against native tool calls (E9: equal accuracy, the grammar sampler costing at least 240 of J's 418 ms mean); the executive's IO round-trip (E10: the tool-response turn invites completion claims, so outcomes stay text); the consult loop (E11: the harness owns the branch, the model chooses within it); one model for both agents (set aside on the Python development stack, where two instances fit; built as the Go deployment's shape, where it ships: the same reads at 1.4 GB less, with a sampled line able to drift past the model's 512-token window); the specialist's limit (S1: a small model cannot continue a REPL read); parity between the Python development stack and the Go embedded runtime on recorded days (30 of 30 fixtures byte-equal, the C ABI's goldens 46/46, 95/95 and 43/43); and a CI job that runs the dog's turns on E2B on a public runner's CPU in five minutes. We argue that the catalog's language — atoms, intents, replies and the rules' own `%%` lines — is the emerging user interface: what the person says, what the device says back and what a developer edits are the same words.

1 Introduction

Robots and appliances run offline, on small hardware, for years. A language model is the natural front for them — people speak, the thing answers — and a small one now fits:

*Sources, the experiments' results, the fixtures and the runtime: https://github.com/metis-lang-dev/metis-agent and its sibling repositories at https://github.com/metis-lang-dev. Correspondence: info@robofortis.com.

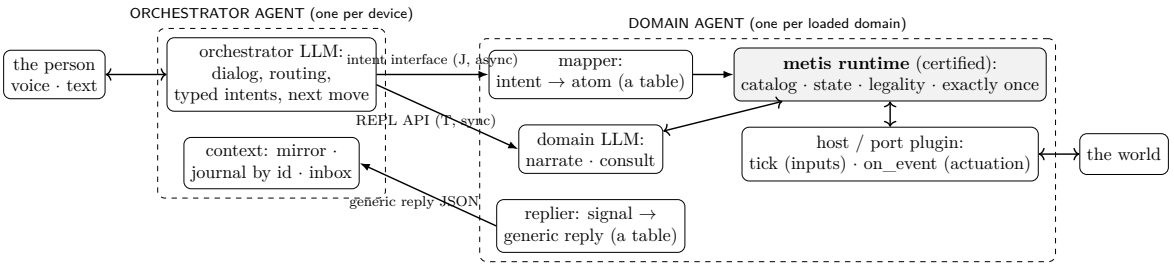


Figure 1: Two agents and one kernel [DESIGN v3, the flowchart]. The runtime inside every domain agent is the same certified interpreter; the orchestrator never touches it directly. The line between the agents is who has to understand the domain: atoms, events, bindings and ports are the domain agent's; the orchestrator knows a domain by its capability line, its verb list and its mirror line.

Gemma 4 E2B, quantized to four bits, is 2136 MiB of GPU memory on a laptop part [MODEL-EXPERIMENTS §1, table 1] and runs on four CPU cores at 17.9 tokens per second [catalog-packages step 1 B report; `.github/workflows/model.yml`]. What a small model cannot do is own the device. It forgets across turns, invents causes when none is given, reads an acknowledgement as completion, and repeats a refused call. Every one of these is measured below. None of them is a reason to use a bigger model; each is a reason to give the model less authority.

The division this paper describes is one of *authority, not speed* [DESIGN "Vocabulary"]. The certified side is a LLP behavior program: a catalog of typed, staged, weighted linear-logic rules which denotes a probability measure over its own runs (the language paper). Its runtime owns state, procedure, probability and safety; every behavior is admitted through gates; an evaluation costs about ten milliseconds; it is always on and never waits for the model. The proposing side is the model: it narrates, maps a request to one typed intent, proposes an explanation for a surprise, describes an escalation. It is fallible by assumption, checked before every effect, and its authority is revocable to zero — the runtime's observe-only mode depends on nothing the model says.

The paper sits in the middle of a three-layer stack. Below it is the language and its toolchain (the OCaml compiler `metisc`, the Python referee `metispy`, a header-only C++ runtime, parity gates between them). Beside it is the catalog — a program as a package: one directory, one manifest, one loader — and the human-gated loop that grows a package by diffs a person approves (the catalog paper). This paper is about what runs on the device: the agents, their interfaces, the deployment shape, and the experiments that chose them.

2 Architecture: two agents, one kernel

The design is stated for two agents on two models; running both on one small model is an optimization, measured in §5.7, that changes none of the interfaces [DESIGN v3 "Architecture"].

Roles. Table 1 is the design's own table of who owns what [DESIGN v3 "Roles"]. The model's column is short on purpose.

The domain agent. One per loaded domain, around one runtime serving one catalog, on one host. Two of its five components are model work, three are tables and code [DESIGN v3 "The domain agent"]:

- the *narrator*: a runtime fragment → one sentence in the person's words, read against the catalog's `%%` comments (single pass, thinking off);

concern	owner	the LLM's part
state over a long run	runtime	reads a mirror of it
step-by-step procedure	domain rules	picks which domain applies
uncertainty	runtime, exact inference	puts the result into words
safety and legality	orchestrator; observe-only on surprise	none; cannot override
when to think	orchestrator signals	wakes when signalled
changing the rules	revision loop, offline, human-approved	describes what went wrong
language and intent	orchestrator	one typed intent per request
understanding a domain	domain agent	narrates it; answers questions about it
judging a surprise	domain agent proposes, runtime checks	suggests which card explains it
the dialog with the person	orchestrator	speaks; asks; keeps the floor

Table 1: Roles [DESIGN v3].

- the *consultant*: a question → a capped loop of REPL verbs → one typed answer (synchronous tool calls);
- the *mapper*: the intent JSON → one command atom from the card's table (`NAVIGATE goal=tag3` → `+cmd_go(tag3)`); open-vocabulary targets resolve to the host's handles; quantities never enter the catalog — they ride with the command in the host's queue and the port actuates with them; a verb with no row is an error. No model: a table, token-exact;
- the *replier*: a runtime signal or an intent's outcome → the generic reply, its `text` from the narrator. No model: a table over the wire's signals;
- the *host*: the port plugin, the catalog author's code: `tick` brings inputs as exogenous rows, `on_event` actuates.

The orchestrator. The person's counterpart and the manager of the domain agents: it briefs, reads reports, decides the next move, asks the person when the world lacks something. It never reasons about why a domain did what it did; it asks the domain agent. Its context is laid out as regions ordered stable to volatile for the KV cache [DESIGN v3 "The orchestrator agent"]: a stable prefix (persona, the two interfaces, the next-move table); a domains block (per loaded domain one capability line and its verb list, never a nested schema); a journal (requests, intents by id and their outcomes, consult answers, surprises); a mirror (one line per domain, overwritten in place); an inbox (the one waking item).

Two interfaces out, no overlap. Table 2 is the contract the experiments of §5 tested and amended.

The generic reply and the clock. Every domain agent answers in one JSON shape: `{domain, kind, id, status, reason, alternatives, text, ref}`, with `kind` in result · event · quiescence · surprise · consult · ports; the orchestrator's next move is a table over `kind` and `status`. Time is beats: the host's tick at quiescence brings the world's inputs as one exogenous row, and a beat may be *scored* — the runtime's likelihood of the host's reading given the beat's start, and per action the program sent, the probability of the world's answer among that action's alternatives (the catalog paper's measure).

The flow of one request. The design's own example [DESIGN v3 "The flow of one request"]; §7 shows it as recorded:

```
person: "go to the kitchen"
orchestrator → intent {"id":"c17","domain":"dog","verb":"NAVIGATE",
                       "args":{"goal":"kitchen"}}
```

	the intent interface	the REPL API
shape	J: one JSON intent under the grammar built from the loaded verb lists	T: native tool calls, synchronous, one per turn
changes the world	yes: `call(domain, verb, args)`; the domain agent maps it to an atom, the runtime fires it under legality, the port actuates	no: it manages and reads
answered	later, by id: `accepted` now; `done` · `refused` (reason, alternatives) · `expired` · `progress` as they happen	now, in the tool response, and *final*: `applied` · `admitted` · `refused` · `failed`; never `accepted`
verbs	the domain's dictionary (the dog: its eleven)	`activate`, `deactivate`; `state`, `ctx`, `events`, `why`, `findings`; `consult`; `reset`, `explain`, `escalate`
the person	—	`ask(question)` is the orchestrator's own, as are `stop` and `cancel`, which fire on the keyword path before any model

Table 2: The orchestrator's two interfaces [DESIGN v3 "Out, two interfaces"]. The world is changed only through intents, which are linear and checked; everything synchronous and read-only or administrative is the REPL API.

```
mapper      → +cmd_go(kitchen)  (the host's tick asserts it as an [env] row)
runtime     → go_closed[P=kitchen] fires
                (the only enabled rule: closed(kitchen) holds)
replier     → {"kind":"result","id":"c17","status":"refused",
               "reason":"the way to it is closed", "text": ...}
orchestrator → no alternatives → REPL consult(dog, "why is the way closed?")
consultant  → why refused(go) → closed(kitchen) →
                {"answer_kind":"missing", "resource":{"atom":"closed(kitchen)",
                      "words":"the kitchen door"}}
orchestrator → ask(person, "The kitchen door is closed.
                          Should I wait, or go somewhere else?")
```

3 What the runtime gives the model

The model never sees the program. It sees what the runtime renders for it, and every rendering is deterministic code, not a model's paraphrase — a lesson E6 taught (§5.2).

The mirror. One line per loaded domain: stage, status, surprise state, the declared beliefs (`#query` marginals projected over the case horizon) and the counts of the atoms the program names. Written by the harness from signals, overwritten in place, at the head of each turn the harness writes [DESIGN v3; CML.md §3]. It is the runtime's state in the atoms' own names, `pred(args)`, and the orchestrator's "what are you doing?" is answered from it (E13 step 3, rule 4).

The journal by id. Requests, intents with their ids and outcomes, consult answers, surprises and how they ended; the model appends and compacts. The `dlg-*` records of the journal are the same on both backends, so one reader (`go/tools/live-reads.py`) scores either [DESIGN "What parity means"].

Surprise records and their triage. The watch checkpoints the program at every beat, reads the host, and prices the reading from the checkpoint. A reading no run reaches is

a surprise: the record carries the program key, the steps, the start, the observed counts, the likelihood, its kind (`unmodeled` or `rare`) and the host's snapshot as data, so it replays against the same world [metispy `kernel/record.py`; RUNTIME-PARITY row 17, 22]. On a surprise the domain goes observe-only; the model's part is to propose a card and move that explain it, and the runtime checks the proposal as a witness over the fired path. A surprise nobody explains is escalated to the people who revise the model, with the record; it is not fixed on the device.

Intents, generated. A domain's `intent.yaml` names the verbs, their meanings, the persona and the examples; from it a generator writes the grammar (`dog.gbnf`), the prompt, the spoken-response list and the corpus (`intent/dog-corpus.tsv`, 54 rows). They are never written by hand [metis-catalog `domains/dog/intent.yaml`; `tools/intents.py`]. The grammar pins the model's turn to one JSON object the mapper's table admits: the certified surface stays token-exact.

The REPL tools, declared once. The REPL's verbs are declared by `metispy` (`repl_api`) and carried by every bundle as `tools.json`; both runtimes render the same declarations, and the orchestrator's decider builds its own three from that data [MODEL-EXPERIMENTS "The decider's declarations" (repl-tools step 3)].

4 Deployment shape

Python develops, Go deploys. `metis-agent/py` is the development stack: the orchestrator and the domain agents run in one process, on one loaded model with three contexts (orchestrator 16k, narrator 8k, consult seat 8k), with the console's backend in the same process; the orchestrator's upper half and the domain agent's lower half are on separate contexts on purpose, so an agent developing a domain can exercise the lower half with the upper half emulated [DESIGN "Two stacks, one behaviour"; agent-console step 2, metis-agent 8d35cfa..3352be8]. `metis-agent/go` is the deployment runtime for the embedded case: a small LLM with a text or voice interface on resource-constrained, offline devices, the cost proportional to the intelligence required [RUNTIME-PARITY § step 2].

The runtime, bound. The device runs the compiled program through a C ABI over the language's header-only C++ runtime (`metis_rt.h`, ABI 2, 23 exported symbols) and a cgo binding (`go/internal/rt`): load the bundle's ground wire in the sidecar's canonical order, bind the weight ports, seed, step, fire, enabled events, do-lines, snapshot/restore, likelihood, hot reload with degrade-to-last-good. Through cgo the language's goldens give IR parity 46/46, ground parity 95/95 and sample parity 43/43; live parity against `metispy`'s REPL on the dog (32/32 rows over 489 events), the plant with its extension (30/30) and the kitchen pair (148/148 and 156/156, ports bound by Go), with snapshot continuity and the recorded surprises' likelihoods equal at 10^{-12} [runtime-binding step 5 report; metis-lang 20096ed, metis-agent 3d99a75]. llama.cpp is a submodule at a pinned commit, built once in CI and cached.

The loader contract. What a runtime loads is an *image* — a composition linked by the language's toolchain, or a lone package linked into a one-member image — and a *plan* naming the members its process holds; it reads the image and nothing else. From the image it takes its members' bundles and the host each binds to (a Go registry must hold that host under that name, else it refuses to load), the clock of each member, the protocol's natures, the port table, the threshold tables its hooks are built from, and the lock it handshakes with. It never

reads the composition file, a manifest, or the source; the Python stack, which develops from source, builds the image through the same link and loads what was written, so there is no second resolver. The beat order inside a tick is part of the contract and the same in every runtime: the specialists first, then the director, which reads this tick's outputs and whose grants land next tick; director first only for a stop. Across processes the same image is loaded on each side with the plan naming the split, and each process sends the other copies of the ports it produces, one message per tick, with the image and plan hashes in every message so a stale side is refused at the first line. The transport is a carrier, not a semantics: sockets today, a robot middleware when a behavior test on a simulator needs it, and the parser on the receiving side is the same atom reader either way. For this paper's shape the split is the dialog on one side — the model, the person's text or voice, the person as the world the behavior cannot command — and the behavior on the other, in one process with every domain it holds; a two-cook kitchen replayed under both plans showed every member's fired rules identical beat for beat, the only difference the first beat's two fetches, whose command crossed the boundary one tick later.

Parity on recorded days. The generic domain agent exists twice — Python over `metispy`, Go over the C ABI — and the bar is logic parity: driven by the same inputs, the same generic replies, the same records [DESIGN "What parity means"]. Without a model, the dog's, the plant's, the kitchen pair's and love_triangle's scripted days are dumped by Python and replayed into Go, the streams diffed as parsed JSON with likelihoods at 10^{-9}: 30 of 30 fixtures byte-equal, re-generated and compared on 2026-10-19 [`go/tools/dagent-parity.sh`; rddl-watch step 1 report]. With a model, the orchestrator ported to Python decodes the same chunks under the same grammars and samplers as the Go one (180/180 scenario replays, 42/42 timeline replays, mutation-checked), and on the model 90 of 114 generations are identical, every first divergence at a token-equal context and none moving a read; the remainder is the two llama.cpp builds' arithmetic, not the port [agent-console step 2 report; DESIGN "What parity means"].

The adapter. One finding is worth its line. The Python voice path was 12.89 ms per token against Go's 7.39 at 5.9k tokens of context, on the same library and the same dialog; the cause was flash attention, which Go left at llama.cpp's default (on for this model on CUDA) and Python had off. With it on, Python is 7.37 ms per token and generates the same greedy text; live, the service went from 18.4 to 9.5 ms per token [llama-pin step 3 report; metis-agent 53ce367].

The package is what both consoles load. A domain reaches the agent as a catalog package — one directory with its manifest, its program, its intents and its goldens — and the agent takes the catalog from the directory it is started in, loading a named package at startup or from the console's catalog widget at any time. The loop that grows a package (the catalog paper) loads the same directory the same way, so a package that passes the catalog's own checks runs unchanged under the loop and under the agent, in Python and in Go. The goldens are generated by the language's reference toolkit and verified by both runtimes; they are the package's contract, not the agent's.

Specialists under a director. A behavior of any size is a composition of specialist packages over interface packages (the language paper): the director hands goals down and reads reports up, each specialist is certified alone against a generic interface stub, and the shared atoms cross as copies at tick boundaries. The agent runs the composition as separate sessions on one kernel, relaying at the boundaries, and a stop from the person is a cancel that reaches

what	where measured	number
E2B Q4_K_M, text path, resident	M-E §1, 8151 MiB card	2136 MiB
E2B Q4 + whisper base.en, co-resident, one process	M-E budget line	2368–2374 MiB
Go service (E2B + the dog's prefix + the orchestrator 8k)	integration step 3	2868–2874 MiB
py domain agents (dog, kitchen, kitchen_shared)	integration step 3	2360–2370 MiB
both resident	integration step 3	5339 MiB (2.7 GB free)
live service, two models → one model (the session's end)	go-domain-agent step 6	3846 → 2436 MiB (−1410)
voice, ms per token at 5.9k context, py off → py AUTO → Go	llama-pin step 3	12.89 → 7.37 → 7.39
voice, ms per token live	llama-pin step 3	18.4 → 9.5
a person turn: move generation, median	integration step 3 (go-scripted)	464 ms
a spoken line: voice generation, median	integration step 3	68 ms
a reply → its line spoken, median	integration step 3	80 ms
the wire, per intent (a tool call waits for its answer: the kitchen's consult, 6.9 s)	integration step 3	0–1 ms
a beat with the narrator	integration step 1 run 2	113 ms
CI, E2B on a public runner's CPU: 3 tests	model.yml, first run	5 min 14 s
llama-bench pp512 / tg64, 4 threads, CI's CPU build (local)	catalog-packages 1B	123 / 17.9 tok/s

Table 3: Deployment numbers. M-E is MODEL-EXPERIMENTS.md; the step reports are the subjects' committed reports.

every active member within one tick. Nothing the model says can hold a resource: grants are the director's, and a member's command-bearing rules read their grant.

The console. The person meets the agent through a web page: fsttm-web's dialog FSM (a finite-state turn-taking model keeps the floor; the models are senses and the proposer, never the turn-taker), attention and speech, over one websocket protocol both backends serve whole [DESIGN "The process shape"; WIRE.md]. The agent console shows every spoken line labelled by its source, the domains, each domain's REPL log, the world's ports and the journal live.

5 Experiments that fixed the design

Each experiment was pre-registered with its reads and a decision rule, run by one implementer, and ruled by the architect. They are held in one document, `docs/MODEL-EXPERIMENTS.md`, and the numbers below are its, verbatim. Table 5 is the summary.

5.1 The model switch (E1–E3)

The question was whether Gemma 4 E2B or E4B could replace gemma3:4b as the intent model, within an 8 GB GPU budget shared with ASR. Same harness, same 54-turn dog corpus (34 prompt rows, 20 held-out), same adapter [M-E §1, table 1; results/e1-p2ex].

The ruling: E2B Q4_K_M, thinking off [M-E "Results and recommendation"; metis-agent 875c219]. E4B bought nothing and cost about 2 GB; thinking bought nothing on intent and 1.3–1.6 s

model	VRAM	turn p50 / p90	intent	held-out	narrate leak
gemma3-4b Q4 (before)	3556 MiB	506 / 833 ms	52/54	16/17	44/54
E2B Q4	**2136 MiB**	**458 / 722 ms**	**54/54**	**17/17**	0/54
E2B Q5	2374 MiB	466 / 742 ms	54/54	17/17	0/54
E4B Q4	3810 MiB	603 / 960 ms	52/54	15/17	0/54
E4B Q5	4292 MiB	637 / 992 ms	53/54	16/17	0/54

Table 4: E1, the intent model on the text path [M-E §1]. "Narrate leak" is the bare single-pass path, not the harness's; through the harness both models narrate cleanly.

per turn. On narration of 24 failed or refused outcomes, both models pass the guard 24/24, but they differ in fidelity: gemma3 invents causes when none is given ("because my joints are stiff"); E2B states the given reason, formulaically, at 108 ms median against 165 [M-E §2]. For ASR, whisper base.en with a command-vocabulary initial prompt gives 1.9% WER on the rendered corpus and 53/54 intents after transcription, in 409 MiB; the streaming Nemotron candidate gave 9.6% and does not endpoint by itself [M-E §3, E2]. The heard path (Gemma 4's native audio into the same grammar) matches the pipeline on accuracy and latency but costs about 0.8 GB more and leaves no transcript, so it is not recommended yet [M-E §4, E3].

5.2 E6: does a spliced fact steer the continuation?

Under the stream idea (CML.md): a runtime result formed into a short text piece and inserted into an ongoing context — does the model reason from it? Twelve real runtime facts from the kitchen fixtures, three renderers, three insertion forms, three distances [M-E E6/E6b; results/e6]. The finding: inline at distance zero, yes, with plain words — the plain-word template T2 reaches 0.92 (thinking off) and 1.00 (thinking on), while the model's own zero-context narration of the raw line loses everywhere (0.25–0.42). A fact does not carry across turns reliably (0.42–0.75 at six turns), the newer of two values does not win reliably (0.75 newer / 0.08 older), and a third of all answers open by copying the spliced line back; up to 0.19 of answers under some forms *forge* a fragment in the runtime's syntax with a fact the runtime never sent. The rulings: the renderer stays deterministic; standing facts go in the mirror, overwritten in place; compaction deletes a superseded fact; and a model output carrying the fragment syntax is refused, never spoken [M-E E6 "What this says"; metis-agent 9eba55d].

5.3 E9: native tool calls against the JSON two-pass

Gemma 4 is trained for function calling; the harness maps an utterance to JSON under a grammar, then continues with a bare "Spoken response:" cue. Does the native call beat the JSON on the corpus we have, and what does the grammar cost? Five arms on E2B [M-E E9 findings; results/e9].

arm	class	JSON identical	ms to intent p50	prefix tokens
J (the JSON two-pass, control)	54/54	54/54	416	1491
J-free (no grammar)	54/54	54/54	175	1491
T (native call, grammar-pinned)	54/54	52/54	531	3481
T-free (native call, unpinned)	53/54	52/54	154	3481
T-noex (no examples)	50/54	32/54	493	2461

Class ties, 54/54; the tool call loses two rows on arguments and is 112 ms slower from its larger grammar; without the 40 examples both class and arguments fall. The ruling: J stays

[M-E E9 ruling]. The finding to act on is latency, not format: the grammar sampler costs at least 240 of J's 418 ms mean (p50 416), and unpinned the model writes a well-formed, correct JSON on 54/54 rows in at most 178 ms — free-then-verify (E9b) is the proposal that acts on it. The certified surface stays grammar-pinned because nothing unvalidated may reach the dispatcher; these numbers are the price of that.

5.4 E10: the executive's IO round-trip

E9 compared classifiers. E10 tested the formats as an *API*: typed calls down, responses the model consumes up, sync and async on the same turn type, over a synthetic corpus of 282 scripted dialogs with injected responses and signals, scored by rules [M-E E10 run; results/e10]. The tool-call arm failed the parity bar (class 52/54, 3 leaks, 4 completion claims against `accepted`) and won none of the deciding groups; it won only interleaving (9/10 against 4/10) because its response carried the call's id. The ruling: JSON text stays in both directions [M-E E10 ruling, user confirmed]. What the three tool-call experiments taught together: accuracy does not depend on the format (the rows a small model gets right are decided by the examples it is shown); declarations cost context the small model pays for (5.2k against 2.8k prefix tokens, two held-out rows); the response turn carries a trained prior — *call, result, done* — so every arm that saw a result turn reported a task complete on some acceptances and the JSON path never did; and identity in the outcome, not the format, prevents cross-wiring. E10b showed the hand-rendered declarations are the GGUF template's rendering byte for byte (17 of 17), and that E4B does not close the gap [M-E E10b ruling].

5.5 E11: the consult loop

After a refusal or a failure, does E2B make the right next move: call an offered alternative, consult, explain, ask, reset or escalate, and never repeat the refused call? Sixty-six items in five groups, reads pre-registered [M-E E11; s-a0477202]. Run 1, with a generic cue, failed three of four reads: the model consults when alternatives are offered, escalates when it should consult, repeats a consult when the same refusal returns. Run 2 changed the harness, not the prompt: the cue names the branch the harness already knows (alternatives offered · none · a consult answered · the same refusal again), the next-move grammar is restricted to that branch's moves, the explanation line is templated. Then G1 12/12, G2 11/12, G3 23/24, G4 6/6, the ask on every missing-resource item and nowhere else, 0 repeats, 0 leaks and claims over 66 lines. Adopted [M-E E11 ruling]. What the two runs settled is the division of labour this paper is about: the harness owns the branch; the model chooses within it and speaks.

5.6 E13 and integration: the dialog under control

E13 moved the pieces into the Go orchestrator and measured it against emulated domains — the dog with live tasks, the kitchen sending orders and signals unasked — on 14 timelines × 3 seeds with bars on ids kept straight, the floor honoured, the person's answer bound to its question, zero signals read as commands, speech clean [M-E E13 steps 1–3]. Three runs and three rulings later every read held: an outcome's line names its own task 45/45, the opener's domain and record carried 24/24, no generation on an event or a notice, yes → the retry 12/12, no → no world-changing intent 3/3, a new command closes an open question 3/3, every intent allowed 78/78, 0 leaks, 0 claims, 0 ids aloud [M-E E13 step 3 run 3b ruling]. Nine harness rules came out of it, all in the design: the bound answer is classified, not moved; the intent pass runs first on a bound turn; the offered verb on a busy question is yes; the outcome turn leads with the task's words; the ask check takes the resource's key; an information turn speaks from the mirror; no intent speaks at the person's turn — the first

reply speaks. The real dog behind the orchestrator then held the same bars (R1a 45/45, R4 78/78) after five more rules the real world exposed [M-E Integration step 1 run 2 ruling; s-f82a578f]. Two further rulings from the live measure: the orchestrator's dialog room went to 16k tokens (+106 MiB), and the surprise's consult runs off the orchestrator's turn so a person is never queued behind it — a dog command typed one second into a 7.5 s consult is dispatched in 596–687 ms and its first line spoken within the beat [integration step 3; agent-console step 2 §5]. Time and date answer from the clock, never from the model.

5.7 E12: one model for both agents

Two decisions, in order. **First, at integration, the single-model optimization was set aside as a task on the Python development stack**, for three reasons: two E2B instances take 5.3 of 8.1 GB and VRAM does not bind there; one model would serialize the kitchen consult (6.9 s) with the orchestrator's own generations; and the one-process shape makes weight-sharing nearly free later [integration step 3 "E12"]. **Second, it was built as the Go deployment's shape** and measured on the same reads [go-domain-agent step 6; metis-agent 60ccbe0, 7074f98]: the orchestrator and every narrator are roles on one loaded context, each with its own budget, prefix, tape and sampling, swapped by KV state save/load (204 swaps in a 14-timeline replay, about 24 ms each). Every read is identical to the two-model run; the dispatched intents are equal 7/7 live; the live service's VRAM falls from 3846 to 2436 MiB at the session's end (−1410; 3792 to 2432 at its start). One finding, with its cause established: past Gemma 4's 512-token sliding window a restored KV is exact in content but laid out differently, the attention's fp16 accumulation moves, and a *sampled* line can take another word — greedy moves keep their argmax, so no read and no dispatch moves.

So, plainly: the Python stack runs two model instances, because it is for development and the two halves must be exercised apart; the Go runtime runs one model, because it is for deployment and the footprint is the point. Byte-exact one-model serving, which needs a layout-preserving KV restore, is the follow-up.

5.8 S1: the specialist's limit

Can a small model be the kernel's user — continue a REPL read, call `why`, `necessity`, `sufficiency` on what it sees and answer faithfully? Twenty items, two reads (reached, faithful), four Gemma seats and Claude as the baseline [SPECIALIST §5b; metis-agent 94cbaea, df1db14]. Reached / faithful of 20: E2B off 10 / 1; E2B with a 256-token thinking cap 16 / 6; E4B off 16 / 8; E4B cap 19 / 7; Claude 20 / 15 (18 by hand). Read 1 holds for E2B with the cap and for E4B; read 2 fails on every Gemma seat, and necessity is where the small model is furthest behind on every seat (Claude 5/5, Gemma ≤ 2). Under the pre-registered rule the specialist line does not continue on the small model as it stands. About half the misses were an argument the kernel could not hold — `state` printed atoms grouped by predicate while the instruments take `pred(args)`, a form no output the model saw ever printed — which is the class of error every E-series found and fixed on the harness side; S1b re-registered that one form change (`metispy` 11c9236) and the bars stand. The consultant stays in the domain agent for the restricted, measured use it has: a surprise with cards, kinds and moves pinned, where the kind is computed from the host's complementary pairs and not chosen by the model at all [DESIGN v3 "The domain agent", system prompt part 2].

5.9 E7/E8: retrieval

Does structural retrieval over the runtime's fragments beat flat mixing of the last pieces of each domain's tape? Run 2: attribution S 0.80 = F 0.80, tokens S 246 < F 254, invented 0

everywhere; the display names, not the retriever, are what made the model attribute a fact to its domain (three questions went 0/3 → 3/3 in every arm) [RAG §10, metispy bf52dbd]. The read: flat mixing suffices until a tree exists; the RAG stays an experiment. What stands in the harness: fragments tagged with the domain's display name in the person's words, one tape per domain on one clock, cross-domain moves naming their target. Triage (which card explains a surprise) was measured apart: the arms without the tape did as well as with it (B 1.00, C 1.00 against A 0.60), so the tape was dropped from triage and the decider runs on the surprise and the cards alone.

5.10 CI on a public runner

The model path — the two-pass intent, the tape kept through turns, two roles on one context — had no CI at all: its tests skip without weights. A workflow of its own now runs E2B Q4_K_M on CPU on a public runner: the weights pinned by revision and sha256, public and ungated under Apache-2.0, cached; llama.cpp built once with CI's flags; three Go tests on the dog's prompt and grammar, with a skipped load counted as a failure; daily, on dispatch, and on a push that touches the adapter, the agent or the llama.cpp pin — never a required check [`.github/workflows/model.yml`; metis-agent 737c4ac, 896f1cb]. Locally with CI's build on four cores: prompt evaluation 123 tokens/s, generation 17.9, one dog verb turn 1.66 s mean, the 54-row corpus 54/54; the first run on GitHub took 5 min 14 s and passed [catalog-packages step 1 B report]. The splice test is left out: on CPU, a spliced KV and the same tape decoded at once part after about fifteen tokens — a numerics limit of an exact-equality test, not a regression, and the same arithmetic the one-model finding of §5.7 met.

6 Language as the emerging user interface

The same words appear three times, and that is the point.

What the person says. The intents are a vocabulary the catalog declares: eleven verbs for the dog, their meanings in a sentence each, the examples the grammar is generated from, the persona — "You are Nina, a quadruped robot dog" — all in `intent.yaml` [metis-catalog `domains/dog/intent.yaml`]. The grammar over the intent schema is a UI contract: whatever is said, what reaches the device is one object the table admits, and the 54-row corpus is the contract's test.

What the device says back. Every rule carries a `%%` line in the device's voice — `%% I sit down.` above `sit [dyn] : cmd_sit * posture(P) * …`, `%% I can't go to {P}: the way to it is closed.` above `go_closed` — and the narrator fills the binding (`{P}` → "the kitchen") from the host's display names. The generic replies are templated from the same lines; the consult's resource words ("open the way to the kitchen") are the host's table's, not the model's, since E2B wrote usable words 1 time in 4 [DESIGN v3 "Open, settled 2026-10-14"]. A refusal names its blocker in the words the person can act on.

What the developer edits. The program, its `%%` lines, the intent file and the extensions are the files the loop proposes as diffs and a person approves (the catalog paper): the words the device will say are reviewed in the same gate as the rule that makes it say them. A change of behaviour is a change of text, never of weights.

	what was asked	result	ruling
E1–E3	E2B/E4B as the proposer; ASR; heard vs read	E2B Q4: 54/54, 458 ms, 2136 MiB; whisper + prompt 1.9% WER; heard = pipeline, +0.8 GB	E2B Q4, thinking off, whisper
E6	a spliced fact steers the continuation?	0.92–1.00 inline at distance 0 in plain words; not across turns; forgeries	deterministic renderer; mirror for standing facts; fragment syntax refused
E9	native tool call vs JSON two-pass	class tie 54/54; args 52 vs 54; +112 ms; grammar $\geq$ 240 ms	J stays; E9b proposed
E10	tool call as the API	T fails parity (52/54, 3 leaks, 4 claims); wins only interleaving	J-text both ways; identity in the outcome
E11	the consult loop	generic cue fails 3/4 reads; branch cue + restricted grammar: all hold	adopted: the harness owns the branch
E13	dialog control vs emulated domains	every bar after 3 runs; 9 harness rules	phase 2 complete
integration	the real dog behind it	every bar after 5 more rules	phase 3 step 1 done
E12	one model for both agents	set aside on the Python stack (two instances fit); built for Go: reads equal, −1410 MiB, a sampled line may drift	two models to develop, one model to deploy
S1	the model as the kernel's user	read 2 fails on every Gemma seat (E4B off 8/20; Claude 15)	the specialist waits for a stronger small model
E7/E8	structural RAG vs flat mixing	S 0.80 = F 0.80; names did it	flat mixing; the RAG an experiment

Table 5: The experiments in one table [MODEL-EXPERIMENTS.md; SPECIALIST.md; RAG.md; the integration and go-domain-agent reports].

Honest limits. The narration is formulaic ("My command to … failed because …"); there are tautologies ("lost sight of you because I couldn't see you"); an adversarial instruction planted in a detail field is repeated, not obeyed [M-E §2]. The runtime's own detail text is what gets repeated, which is a harness concern; the remedy applied is the guard, which turns anything unspeakable into "Okay.".

7 A session, as recorded

Figure 2 is the opening of the dog's recorded day, the fixture the Python and Go domain agents are held equal on [`go/internal/dagent/testdata/days/dog.out.jsonl`, rows `c1`–`c2`]: an intent in, `accepted` now, the rule that fired a beat later with the device's line, a refusal with its reason and alternatives, and the consult that turns the refusal into what the person can do. Every line is the fixture's.

8 Related work

The rules are in the Ceptre lineage of linear-logic programming for interactive worlds (Martens, *Ceptre: a language for modeling generative interactive systems*, AIIDE 2015); what metis adds is the measure over runs and the certified evaluation (the language paper). Turn-taking is the finite-state turn-taking model of Raux and Eskenazi (NAACL-HLT 2009), kept as the dialog engine in front of every model. Constraining a model's output to a

```
→ {"id": "c1", "domain": "dog", "verb": "LOCAL_ACTION", "args": {"action":
"SIT_DOWN"}}
← {"kind": "result", "id": "c1", "status": "accepted", "text": "Accepted."}
  beat 2, event sit[P=standing]
← {"kind": "result", "id": "c1", "status": "done", "text": "I sit down.",
"voice": "line"}
→ {"id": "c2", "domain": "dog", "verb": "NAVIGATE",
   "args": {"goal": {"type": "REGION", "description": "kitchen"}}}
← {"kind": "result", "id": "c2", "status": "accepted", "text": "Accepted."}
  beat 4, event go_closed[P=kitchen]
← {"kind": "result", "id": "c2", "status": "refused", "reason": "the way to it
is closed",
   "text": "I can't go to the kitchen: the way to it is closed.",
   "alternatives": [{"verb": "go", "args": {"place": "door"}},
                    {"verb": "go", "args": {"place": "tag3"}}]}
→ {"tool": "consult", "args": {"question": "why did you not go?"}, "id": "k1"}
← {"kind": "consult", "id": "k1", "answer_kind": "missing",
   "resource": {"atom": "closed(kitchen)", "words": "open the way to the
kitchen"},
   "evidence": ["closed(kitchen) 1", "mapped(kitchen) 1"], "calls": 0}
```

Figure 2: The dog's recorded day, rows c1, c2 and k1 of `dog.out.jsonl` (fields abridged; the person's words for these intents are the corpus rows "sit" and "go to the kitchen"). The consult's answer is computed, not generated: `missing` because the host's pair `open|closed` can be flipped by the person, the words from the host's table; `calls: 0` — no model call was needed.

grammar is llama.cpp's GBNF sampler; §5.3 measures what it costs here. LLM agents that act through tools (the ReAct line, Yao et al., ICLR 2023) put the model in charge of the loop; here the loop is the harness's and the model's authority is a table of narrow, checked jobs (Table 1). Neuro-symbolic planners pair a model with a planner it calls; here the planner side is a probabilistic program whose every evaluation is exact and whose surprise is a measured zero, and the model never calls code — what it may "call" is a catalog domain. On-device models: the budget line of Table 3 is the whole of our claim about size.

9 Limits and what is next

Observe-only is the fallback, and it depends on nothing the model says: a device whose model is wrong, slow or absent still runs its program, refuses what is illegal and records what it cannot explain. The model's authority is revocable to zero by construction.

Not covered here: the user's real recordings (the ASR and heard-path decisions wait for them [M-E "Open items"]); the heard path's projector re-pack; the road world, which now runs as a closed loop over a traffic simulator in the worlds repository and waits for its device shape; and the tuning of the RDDL worlds' programs against their gyms, which the catalog paper's measure now makes possible and this design keeps off the device. The specialist waits for a stronger small model, or a kernel-side form change that S1b measures. Byte-exact one-model serving needs a layout-preserving KV restore. The orchestrator has no journal compaction yet: near the context's end the dialog starts over, which E6 ruled the safe version. Each of these is a subject with a pre-registered read, as everything above was; the record is the repository.

Sources

pointer	document (and commit)
DESIGN	`metis-agent` `docs/DESIGN.md` v3 (2026-10-13; the process shape 2026-10-14, two stacks 2026-10-15; rules 7bc8fd2, c77232b)
M-E	`docs/MODEL-EXPERIMENTS.md`: §1–5 the E1–E4 results (2026-10-11; the switch 875c219), E6/E6b (2026-10-12), E9 (2026-10-13; material later removed in 6fb341f, the record kept), E10/E10b (2026-10-13; s-de466019), E11 (s-a0477202), E13 (s-c9276dee), integration (s-f82a578f), the decider's declarations (repl-tools s-0c6879a1)
SPECIALIST	`docs/SPECIALIST.md` §5b, S1 (metis-agent 94cbaea, df1db14), S1b (`metispy` 11c9236)
RAG	`docs/RAG.md` §9–10 (E7 run 2: `metispy` bf52dbd; bench 06a07ce; scorer 9c30d33)
CML, WIRE	`docs/CML.md` (the regions, §4 the flow numbers), `docs/WIRE.md`
RUNTIME-PARITY	`docs/RUNTIME-PARITY.md` (s-c39746ca, 2026-10-15)
runtime-binding	s-9054df5d step 5 report (metis-lang 20096ed; metis-agent 3d99a75): the C ABI, `go/internal/rt`, the goldens
go-domain-agent	s-a047870a step 6 report (metis-agent 60ccbe0, 7074f98): the one-model shape
agent-console	s-7f3d1088 step 2 report (metis-agent 8d35cfa, 4304c63, 0a13ebe, f9ce9e9, 3352be8): the Python orchestrator and service
llama-pin	s-aeb12858 step 3 report (metis-agent 53ce367): flash attention, the voice profile
integration	s-f82a578f step 3 report (metis-agent fae45b3): VRAM, latency, E12
catalog-packages 1B	s-a2c529d6 step 1 part B report (metis-agent 737c4ac, 896f1cb): `model.yml`, E2B on CPU
parity on days	`go/tools/dagent-parity.sh`, `go/internal/dagent/testdata/days/`; rddl-watch s-cfe94dca step 1 report (the fixtures 30/30, 2026-10-19)
the dog	metis-catalog `domains/dog/`: `dog.llp`, `intent.yaml`, `intent/dog-corpus.tsv`